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| **Fiscal Year:** | FY 2014 | **Task Last Updated:** | FY 10/01/2014 |
| **PI Name:** | Naidu, Mamta Ph.D. | | |
| **Project Title:** | High LET-radiation induced DNA repair mechanisms in Oligodendrocyte progenitor cells (OPC) in vitro and in vivo | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | HUMAN RESEARCH | | |
| **Program/Discipline--Element/Subdiscipline:** | HUMAN RESEARCH--Radiation health | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **SR**:Space Radiation | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders<br>(2) **CNS**:Risk of Acute (In-flight) and Late Central Nervous System Effects from Radiation Exposure | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
| **PI Email:** | mamta.naidu@griresearch.org | **Fax:** | FY |
| **PI Organization Type:** | NON-PROFIT | **Phone:** | (617) 444-3050 |
| **Organization Name:** | GeneSys Research Institute/Tufts University School of Medicine | | |
| **PI Address 1:** | Center for Cancer Systems Biology/Department of Medicine | | |
| **PI Address 2:** | 736 Cambridge Street, CBR 3 | | |
| **PI Web Page:** | | | |
| **City:** | Boston | **State:** | MA |
| **Zip Code:** | 02135 | **Congressional District:** | 7 |
| **Comments:** | NOTE: Previous affiliation was State University of New York-Stony Brook, from October 2011-December 2012. Prior to that, affiliation was Brookhaven National Laboratory, Upton, NY, until September 2011. | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | 2009 Space Radiobiology NNJ09ZSA001N |
| **Start Date:** | 01/23/2013 | **End Date:** | 06/30/2014 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 2 |
| **No. of PhD Candidates:** | | **No. of Master' Degrees:** | |
| **No. of Master's Candidates:** | | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | | **Monitoring Center:** | NASA ARC |
| **Contact Monitor:** | Griko, Yuri | **Contact Phone:** | 650-604-0519 |
| **Contact Email:** | Yuri.V.Griko@nasa.gov | | |
| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: Extended to 6/30/2014 per A. Chu/ARC; original end date was 10/6/2013 (Ed., 11/8/13) | | |
| **Key Personnel Changes/Previous PI:** | | | |
| **COI Name (Institution):** | | | |
| **Grant/Contract No.:** | NNX13AD74G | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | [Ed. note: Grant is a continuation of NNA10DA32I at Brookhaven National Laboratory and NNX11AO89G at SUNY-Stony Brook per PI's move in January 2013 to GeneSys Research Institute. See task, "High LET-radiation induced DNA repair mechanisms in Oligodendrocyte progenitor cells (OPC) in vitro and in vivo" (NNA10DA32I) and "High LET-radiation induced DNA repair mechanisms in Oligodendrocyte progenitor cells (OPC) in vitro and in vivo--NNX11AO89G" for previous reports.]<br>NASA's radiation research program is focused on assuring that humans can travel in space without exceeding the limiting risks from radiation, and on advancing the understanding of response mechanisms to radiation-caused DNA damage. Radiation studies on the CNS have predominantly focused on neurons, with few studies directly addressing the role of glial cells. The focus of our research proposal will be to determine the major DNA repair pathway(s) induced by HZE radiation in glial cells. Oligodendrocytes (OL) are responsible for forming and maintaining myelin, the insulating sheaths of neuronal axons. The damage to OL causes neurological impairment as OL have a slow turnover and are repopulated by resident OL progenitor cells (OPC). Astronauts may encounter repeated exposure to charged particles over the course of their missions such as traveling to Mars and this exposure to low doses of cosmic rays has the potential to prevent astronauts from successfully meeting mission requirements. Base excision repair (BER) is one of the predominant DNA damage repair pathways up regulated in response to oxidative stress by low LET radiation. Also, Apurinic endonuclease 1 (Ape1), a key BER player may be involved in adaptive response by linking repair to oxidative pathways, thus, it becomes important to study its role in fractionated versus single low dose of high LET radiation. As Ape1 is known to affect the G1/S transition in hematopoietic embryoid bodies (EB) cells, it also becomes important to determine if cell cycle dependent changes in Ape1 levels can be correlated to differentiation capacity of CG-4. Thus, our project will determine the effects of high LET radiation on the nuclear and mitochondrial BER in two systems; the central glia-4 (CG-4), a glial progenitor cell line that can differentiate into OL or astrocytes, and in OPC cells isolated from rat spinal cord. The latent CNS effects are due to altered DNA repair resulting in altered OPC differentiation causing neuro-inflammation, inhibition of short term potentiation and decreased cognitive function. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | Our proposed study of the effects of fractionated, low dose versus single low dose of charged particles on glial cells of brain would help us understand the risk to astronauts exposed to chronic radiation exposures and the risk of normal tissues of patients receiving fractionated doses to their brain tumors.<br>Our study will answer a critical question, "Would lower doses of higher Z particles produce more subtle latent changes, for example, changes in CNS cell population type that do not produce externally obvious deficits until months or years after exposure?" The oxidative stress produced by particles affects DNA repair capacity and our studies will show which DNA repair pathways are affected and how they affect the cells' differentiation responses to stress, thereby possibly altering cognitive behaviour, finally leading to better understanding of risks out there for our astronauts on long term missions. |
| **Task Progress:** | 1) Fractionated dose of 1 Gy of gamma rays causes increase in DSBs with an increase in APE1 until two fractions and then shows decrease by third fraction.<br>2) Fractionation of HZE at 0.5 Gy of 28Si causes more senescence than single dose, as compared to fractionation of 250 MeV protons.<br>3) siRNA inhibition of APE1 at 30-50% causes inhibition of CG4 progenitor cell differentiation into OL and astrocytes, which is further increased with 0.5-1Gy of IR, causing most cells to prematurely senesce within 48h.<br>4) Simulated microgravity seem to exacerbate the low dose low LET induced alteration in APE1 and further inhibition of progenitor cell differentiation.<br>5) Fractionated low dose of 300 MeV/n 28Si/ 600 MeV/n 56Fe and 250 MeV protons predominantly induces latent non-targeted tumors (cerebellum, kidneys, lungs, neck, gonads) formation, hemorrhaging in lungs (Si and Fe) and brains (protons) seen with only irradiating the rat thoracic spinal column (4-9 months post exposure) in middle aged rats (9-12 months old). Also we found increase in inflammatory cytokines TNF and NFkB with fractionation, in the heart, blood, and tissues from these rats.<br>6) Fractionation of 0.5 Gy of 28Si causes increased skewing of OPC differentiation, increased alterations in BER and DSB repair.<br>7) 1 Gy of 250 MeV Protons caused radiation damage in spine by augmentation in baseline synaptic strength in Layer II of PFC, while 0.5 Gy of 600 MeV/n 56Fe radiation only at spinal cord caused augmentation in short-term potentiation in rat brain neurons. There was decrease in dendritic spines number as well as density with fractionation of HZE. |
| **Bibliography Type:** | Description: (Last Updated: 11/01/2023) |
| **Abstracts for Journals and Proceedings** | Peluso MP, Kumar MA, Chaudhary P, Wang Z, Wang Y, Lamont C, Madu A, Weremowicz J, Hlatky L, Naidu M. "Fractionated protons and HZE radiation to rat spinal cords increases base excision repair, induces demyelination, neuroinflammation in brain, cognitive deficits and non-targeted tumor induction." Poster presented at 2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014.<br>2014 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 12-13, 2014. http://www.hou.usra.edu/meetings/hrp2014/pdf/3001.pdf , Feb-2014 |
| **Articles in Peer-reviewed Journals** | Suresh Kumar MA, Peluso M, Chaudhary P, Dhawan J, Beheshti A, Manickam K, Thapar U, Pena L, Natarajan M, Hlatky L, Demple B, Naidu M. "Fractionated radiation exposure of rat spinal cords leads to latent neuro-inflammation in brain, cognitive deficits, and alterations in apurinic endonuclease 1." PLoS One. 2015 Jul 24;10(7):e0133016. eCollection 2015. http://dx.doi.org/10.1371/journal.pone.0133016 ; PubMed PMID: 26208353; PubMed Central PMCID: PMC4514622 , Jul-2015 |

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| **Articles in Peer-reviewed Journals** | Sridharan DM, Asaithamby A, Blattnig SR, Costes SV, Doetsch PW, Dynan WS, Hahnfeldt P, Hlatky L, Kidane Y, Kronenberg A, Naidu MD, Peterson LE, Plante I, Ponomarev AL, Saha J, Snijders AM, Srinivasan K, Tang J, Werner E, Pluth JM. "Evaluating biomarkers to model cancer risk post cosmic ray exposure." Life Sci Space Res (Amst). 2016 Jun:9:19-47. https://doi.org/10.1016/j.lssr.2016.05.004 ; PMID: 27345199; PMCID: PMC5613937 , Jun-2016 |
| **Articles in Peer-reviewed Journals** | Gao X, McDonald JT, Naidu M, Hahnfeldt P, Hlatky L. "A proposed quantitative index for assessing the potential contribution of reprogramming to cancer stem cell kinetics." Stem Cells Int. 2014;2014:249309. Epub 2014 May 12. http://dx.doi.org/10.1155/2014/249309 ; PubMed PMID: 24955094; PubMed Central PMCID: PMC4052692 , May-2014 |
| **Articles in Peer-reviewed Journals** | Chowdhury SM, Surhland C, Sanchez Z, Chaudhary P, Suresh Kumar MA, Lee S, Peña LA, Waring M, Sitharaman B, Naidu M. "Graphene nanoribbons as a drug delivery agent for lucanthone mediated therapy of glioblastoma multiforme." Nanomedicine: Nanotechnology, Biology, and Medicine. 2014 Aug 13. [Epub ahead of print] http://dx.doi.org/10.1016/j.nano.2014.08.001 ; PubMed PMID: 25131339 , Aug-2014 |